

Wine Cultivar Clustering: An Unsupervised Learning Study

Comparing K-Means and Hierarchical Clustering, with PCA and t-SNE Dimensionality Reduction

1. Introduction

This report summarizes an unsupervised learning study on the UCI/sklearn Wine dataset, which contains 13 continuous chemical measurements (alcohol, malic acid, color intensity, flavonoids, proline, and others) for 178 wine samples drawn from three cultivars grown in the same region of Italy. The objective was to determine whether unsupervised clustering, using only the chemical measurements, could recover the true cultivar structure without ever having access to the cultivar labels themselves. The true labels were withheld during clustering and used only afterward, purely to validate the results.

2. Methodology

Preprocessing: the dataset contained no missing values and no categorical variables, so the only required preprocessing step was feature standardization (zero mean, unit variance), which is essential since both K-Means and PCA are sensitive to feature scale — without it, high-magnitude features such as *proline* (values up to ~1680) would dominate distance calculations purely due to scale, not actual importance.

Clustering: two clustering algorithms were applied to the standardized data: K-Means and Agglomerative (Hierarchical) Clustering with Ward linkage. The optimal number of clusters was determined using both the Elbow method (inertia vs. k) and Silhouette score (a measure of how well-separated clusters is, ranging from -1 to +1), evaluated across $k = 2$ –10.

Dimensionality reduction: Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE) were both used to project the 13-dimensional standardized data into 2 dimensions for visualization. PCA preserves global variance via a linear transformation; t-SNE instead prioritizes preserving local neighborhood structure, often producing tighter visual cluster separation at the cost of interpretable axes.

3. Results

Optimal k selection: both the Elbow method and Silhouette score independently pointed to $k = 3$ as optimal. The Silhouette score peaked sharply at $k = 3$ (0.2849), dropping at $k = 4$ –5 before a smaller secondary local maximum at $k = 6$, with a clear declining trend afterward. Notably, $k = 3$ also matches the true number of wine cultivars — a result the clustering algorithm reached with no access to that label information.

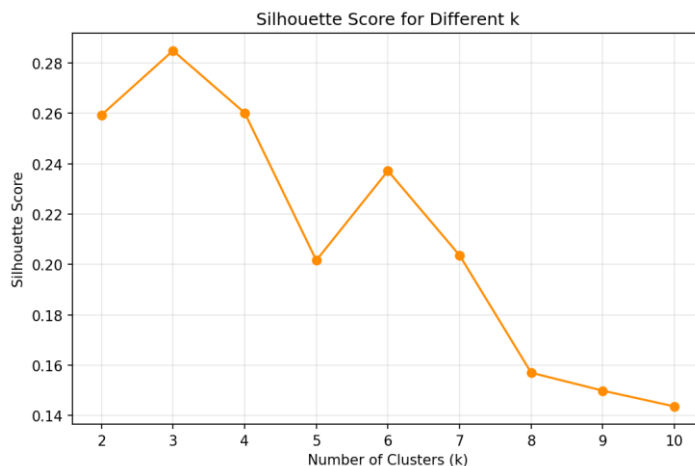


Figure 1. Silhouette score peaks clearly at $k = 3$, matching the dataset's true cultivar count.

Method	Silhouette Score	Cluster Sizes (0 / 1 / 2)
K-Means (k=3)	0.2849	65 / 51 / 62
Hierarchical (k=3, Ward)	0.2774	58 / 56 / 64

Clustering comparison: K-Means slightly outperformed Hierarchical Clustering on Silhouette score (0.2849 vs. 0.2774), though both produced comparable cluster sizes. Agreement between two fundamentally different algorithms, one centroid-

based and iterative, one bottom-up and distance-based — strengthens confidence that the 3-cluster structure reflects genuine patterns in the chemical data rather than an artifact of either specific method

Validation against true labels: cross-tabulating K-Means cluster assignments against the withheld true cultivar labels showed near-perfect recovery: Cluster 0 mapped perfectly to Cultivar 1 (65/65 points, zero misassignment), Cluster 2 mapped almost entirely to Cultivar 0 (59/62), and Cluster 1 mapped almost entirely to Cultivar 2 (48/51). Only 6 of 178 points (3.4%) were assigned to a cluster that did not match their true cultivar, a strong result given that the clustering algorithm never saw the labels

Dimensionality reduction: PCA's first two components captured **55.41%** of the total variance across the original 13 features — a substantial amount for just two dimensions, explaining why the 2D PCA visualization already separates the three groups reasonably well. t-SNE produced noticeably tighter, more visually distinct clusters than PCA (Figures 2–3), consistent with its design goal of preserving local neighborhood structure rather than global variance; the trade-off is that t-SNE's axes carry no direct interpretable meaning, unlike PCA's variance-ranked components

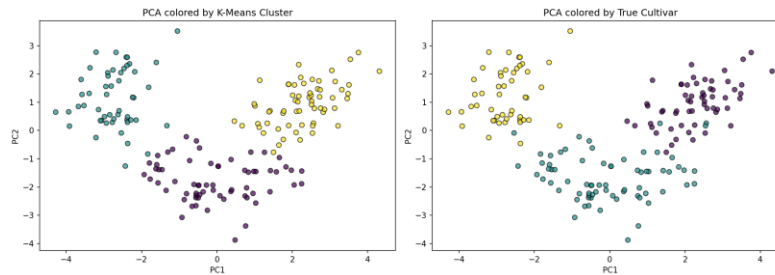


Figure 2. PCA projection colored by K-Means cluster (left) versus true cultivar (right) — the near-identical groupings visually confirm the clustering recovered genuine structure.

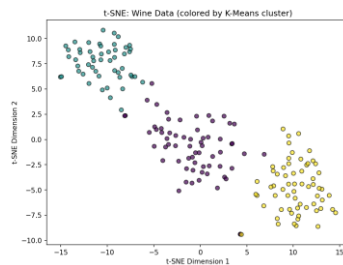


Figure 3. t-SNE projection colored by K-Means cluster, showing tighter separation than the PCA view.

4. Deployment & Monitoring

Hypothetical scenario: a wine producer could deploy the K-Means model to classify incoming samples by likely cultivar from lab chemistry alone, reducing reliance on manual review. The fitted scaler and model would be packaged together and served behind a lightweight REST API, since predictions require the exact scaling parameters learned during training

Key challenges: latency and scalability are minor concerns given K-Means' cheap inference cost. The larger risk is **data drift** — if sourcing or processing changes over time, each cultivar's chemical fingerprint may shift, and K-Means will still confidently assign new points to the nearest cluster even when they no longer belong, with no built-in warning

Monitoring strategy: track feature-distribution drift (e.g., Kolmogorov–Smirnov tests); monitor Silhouette scores on new points for degradation; retrain on a fixed cadence or upon drift, validating against the existing model before promotion; and periodically have assignments reviewed by a human expert

5. Conclusion

Both K-Means and Hierarchical Clustering independently identified three natural groupings, matching the true cultivar count without using any label information. K-Means scored marginally higher on Silhouette (0.2849 vs. 0.2774) and recovered the true structure with 96.6% agreement. PCA explained over half the variance in two dimensions, while t-SNE gave the cleanest visual separation, together showing real structure can be recovered from purely unsupervised analysis alone